Which Idea Should We Build?

Comparing five Track 2 (Evidence Navigator) ideas for the OncoSys Cancer AI hackathon · November 12, 2026

Short answer: none of the five ideas should be built exactly as written. The strongest choice is a **combination**: the **Research Claim Auditor** with robustness checks (Idea 4) as the engine, and the **Claim Ladder** from Ceiling (Idea 2) as the main screen. It is the only option that has an answer key on judging day, actually rechecks the data, and has a screen a judge understands in seconds.

1. What decides who wins

From the rules, product briefs and envelope instructions in the challenge repo:

- **Judges test your tool on a hidden input.** On Saturday each team opens a sealed envelope that holds **one input file per featured product**. An idea that is not a featured product gets no proper test input.
- **Two Track 2 products have an answer key:** Repair Review Desk and Research Claim Auditor. The organizers' checker compares your output with the correct answers, and judges see that comparison while scoring.
- **"A finished workflow beats an ambitious one that doesn't run."** (from the rules)
- **No clinical claims:** no diagnosis, treatment advice or risk scores.

2. What we checked in the real data

We downloaded the frozen TCGA-BRCA release and tested whether the ideas can actually be built:

- **The fields exist.** Sex (1,229 female, 13 male, 5 unknown), the hospital code (40 sites), two different subtype labels, and two survival measures (OS and PFI). So Ideas 3-5 can be built.
- **The practice report contains a wrong claim, and checking it several ways catches it.** The report says "about 62% of primary tumors are Luminal A". The real figure is **49.9% (421 of 844)** with one subtype label and **44.3% (of 522)** with the other, and it barely changes if only women are counted. Under every reasonable choice it is **never 62%**, so the claim is clearly contradicted.
- **Hidden trap:** the two subtype columns spell names differently ("LumA" vs "Luminal A"). A tool that misses this reports 0%. This is the kind of problem judges plant.
- The practice report's "535 of 1,101 primary tumors have all five data layers" matches the data (**supported**). Its claim that high ER-alpha patients "responded best to hormone therapy" cannot be checked, because the data has no treatment-response records (**insufficient evidence**).

Choice	Luminal A share	Samples counted
RNA-seq subtype call, all primary tumors	49.9%	421 of 844
RNA-seq subtype call, women only	49.9%	833 subtyped
2012 study subtype label, all primary tumors	44.3%	522 subtyped
What the report claimed	about 62%	not stated

3. The five ideas side by side

Idea 1: Repair Review Desk · Idea 2: Ceiling (claim ladder) · Idea 3: re-run the analysis many ways · Idea 4: Claim Auditor + many ways + skeptic tests · Idea 5: many AI agents. The highlighted column is the recommended base.

	Idea 1 Repair Desk	Idea 2 Ceiling	Idea 3 Many ways	Idea 4 Auditor+	Idea 5 AI agents
Hidden test file + answer key	Yes	No	No	Yes	No
Original / memorable	Medium (many teams may pick it)	High	High	High	High
Rechecks the data itself	Yes	No: only reads the finding list	Yes	Yes	Partly (AI-driven)
Reliable in a live demo	Yes	Yes	Yes	Yes, if scope is trimmed	No: slow, answers change between runs
Buildable in 2 days	Easy	Depends on files we don't have	Medium	Medium: must cut scope	Hard
Medical-safety risk	Low	Low	Survival claims sound clinical	Low if survival stays minor	Medium
Overall	Safe, solid	Great screen, weak engine	Good engine, no test input	**BEST BASE**	Too risky

Main weakness of each idea

- **Idea 1, Repair Review Desk:** reliable and has an answer key, but it is the obvious choice. The decisions are only approve or reject on a few repairs, so it is less impressive.
- **Idea 2, Ceiling:** the best *screen* of the five. Crossing out words the evidence can't support is understood in seconds. But it only reads the finding records and never recalculates anything (it trusts the file), those records aren't in the repo, its rules for crossing out words are invented by us, and it has no answer key.
- **Idea 3, re-run many ways:** a strong check, but no product is attached, so judges have no test file for it. It is also built around survival claims, while the practice report's claims are about counts and percentages.
- **Idea 4, Auditor + many ways + skeptic:** the right engine with the right wrapper. The risk is doing too much: 8 skeptic tests and 64 versions of every claim won't be finished by Saturday.
- **Idea 5, many AI agents:** live agents give different answers each run and are slow. In a live test that is a gamble. Idea 4 keeps the concept but does it with ordinary code.

4. The recommended combination

Research Claim Auditor (Idea 4) + Claim Ladder (Idea 2). In basic words: you give the tool a report. For each sentence it shows:

- **A verdict:** supported, contradicted, or the data can't tell.
- **How solid the verdict is:** "holds no matter how you count", or "only true if you count one specific way".
- **The strongest honest version of the sentence**, with the over-strong words crossed out. For example: ~~about 62% of~~ **49.9% (421 of 844) of** primary tumors with an RNA subtype call were Luminal A.
- **One decision for the whole report:** Proceed, Fix, Narrow or Stop, plus who must sign off, and an `audit.json` file for the organizers' checker.

How it scores on the Track 2 rubric

Criterion	Points	How the combination covers it
Decision usefulness	20	One overall decision, plus a corrected sentence that can actually be published
Understanding the evidence	20	Click any number to see the samples, the file and the code behind it
Role fit	15	Leader sees the decision; researcher sees how solid it is; engineer sees the code; reviewer sees the sign-off list
Safety and uncertainty	15	Solidity tags, "data can't tell" verdicts, crossed-out over-strong words, research-only banner
Usability and accessibility	15	The crossed-out sentence is understood in seconds; plain words; keyboard use
Working tool and setup	10	Plain Python app, works without an AI key
Communication	5	"The report said 62%. It's really 50%, however you count it."

The answer key also confirms the tool works: no other option gets both an objective score and a memorable screen.

What to cut so it gets finished

- **Keep 3 "count it differently" choices** (which subtype label; women only or everyone; primary tumors or all samples). That is 8 versions, not 64.
- **Keep 2 skeptic tests:** shuffle the labels, and leave one hospital out.
- **Make the AI optional:** it can only help pull sentences out of the report. It never sets a verdict.
- **Cut:** survival models, live AI agents, slide images.
- **Safety net:** if time runs out, the Claim Auditor part on its own still runs and still gets the answer-key score, so the downside is small.

5. Honest risks

Risk	How to reduce it
The judges' report has claim types we didn't prepare for	Build ready-made checks for common types: sample counts, percentages, group comparisons, and "data can't answer" (treatment, cause, prediction)
We haven't seen the exact audit.json the checker expects	Validate early and often with format/validate.py
Pulling claims out of a report is unreliable	AI is optional; fall back to splitting sentences and letting the user tick which ones are claims
Setup breaks on a judge's machine	Use the Friday repo swap with the paired team; pin package versions
A teammate wrote Ceiling (Idea 2)	Its best part becomes the main screen: a real merge, not a rejection

6. Build order (Thursday to Saturday 2:15 PM code freeze)

When	Goal
Thu, Nov 12	Checks for the 3 practice-report claims working on real data. Post as Checkpoint 1.
Fri, Nov 13	Claim Ladder screen, click-to-proof, solidity tags, audit.json export. Repo swap with the paired team; fix whatever they trip on.
Sat, Nov 14 (morning)	Role views, overall decision and sign-off record, accessibility pass, README, demo video. Rehearse with the practice envelope.